

MoELoRA v2:

Per-Tensor Mixture-of-Experts Routing over a B-LoRA Style Zoo

Progress Report — February 2026

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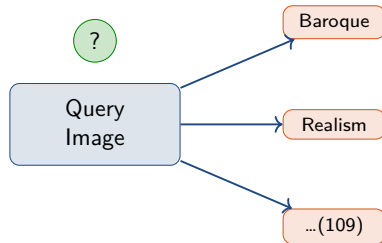
Motivation & B-LoRA Background

Problem: Zero-shot Style Routing

Given **any** query image $\Rightarrow$ generate in same artistic style via SDXL, without fine-tuning.

- B-LoRA zoo: **109 single-style experts** (Baroque, Realism, ...)
- Real queries span an **open vocabulary** of styles
- No clean label $\Rightarrow$ cannot just pick expert i

$\Rightarrow$ **Which experts? How much? Blended how?**



Which experts to activate? How to blend?

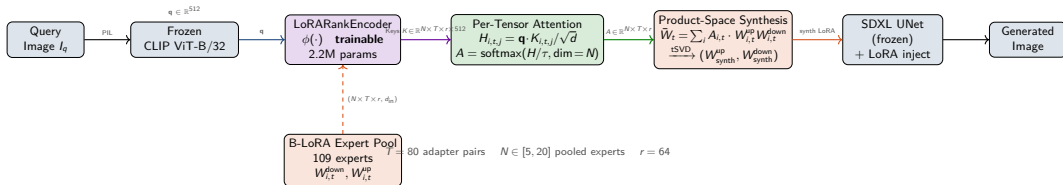
B-LoRA recap

- Target: SDXL style block
up_blocks.0.attentions.1
- 80 adapter pairs (W_t^{down} , W_t^{up}), rank $r = 64$
- Trained on ~ 30 images / style
- Inference: $W = W + \alpha \cdot W^{\text{up}} W^{\text{down}}$

Zoo: 109 experts, 20 styles

Baroque, Impressionism, Cubism, Expressionism,
Realism, Romanticism, ...(WikiArt)
Each: 80 $\times$ LoRA pairs, rank 64

System Overview



LoRARankEncoder $\phi(\cdot)$

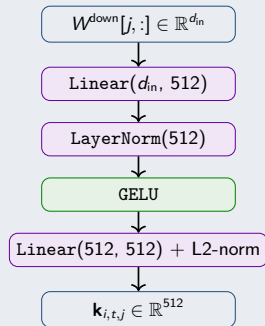
Architecture

Shared **point-wise MLP** over rank positions.

For each rank position $j \in [0, r)$:

$$\mathbf{k}_{i,t,j} = \phi(W_{i,t}^{\text{down}}[j, :]) \in \mathbb{R}^{d_{\text{clip}}}$$

Two separate heads for the two d_{in} sizes:



Why this design?

- Each LoRA row $W^{\text{down}}[j, :]$ is a **rank-direction vector** in weight space
- Encoding each independently $\Rightarrow$ **per-rank routing**
- Batched as a Conv1d $\Rightarrow$ efficient: *only 2 kernel launches* for all $N \times T$ tensors grouped by d_{in}

Parameter count

Module	Params
head_1280	659K
head_2048	1,573K
Total	2.23M

vs. 17.3M in v1.0 RoutingMLP

Forward pass (math)

1. CLIP query encoding:

$$\mathbf{q} = \text{CLIP}(I_q) \in \mathbb{R}^{512}$$

2. Key generation per rank:

$$K_{i,t,j} = \phi(W_{i,t}^{\text{down}}[j, :]) \in \mathbb{R}^{512}$$

3. Dot-product attention score per rank:

$$H_{i,t,j} = \frac{\mathbf{q} \cdot K_{i,t,j}}{\sqrt{512}} \in \mathbb{R}$$

4. Cross-expert softmax (over i):

$$A_{i,t,j} = \frac{\exp(H_{i,t,j}/\tau)}{\sum_{i'} \exp(H_{i',t,j}/\tau)}$$

Result: $A \in \mathbb{R}^{N \times T \times r}$

N experts, $T = 80$ tensors, $r = 64$ rank positions

Key properties

- **Per-tensor**: layer t can route differently from layer t'
- **Per-rank**: rank direction j can prefer different experts
- Attention is **normalised across experts** (dim=0), not across tensors
- Temperature $\tau = 1.0$ at training; $\tau = 0.005$ for sharp inference

Optional: Top- k sparsification

Keep top- k experts per (t, j) , zero others, renormalise.

$k = 1 \Rightarrow$ hard routing (greediest).

$k = \text{None} \Rightarrow$ soft mixing of all N experts.

The Cross-Term Problem in Naïve LoRA Averaging

Naïve averaging (v1.0 — WRONG)

$$\bar{W}^{\text{up}} = \sum_i A_i W_i^{\text{up}}, \quad \bar{W}^{\text{down}} = \sum_i A_i W_i^{\text{down}}$$

$$\Delta W_{\text{naive}} = \bar{W}^{\text{up}} \bar{W}^{\text{down}}$$

Expanding:

$$= \underbrace{\sum_i A_i^2 W_i^{\text{up}} W_i^{\text{down}}}_{\text{wanted: style signal}} + \underbrace{\sum_{i \neq i'} A_i A_{i'} W_i^{\text{up}} W_{i'}^{\text{down}}}_{O(N^2) \text{ cross-terms}}$$

Cross-terms are independent style-component products $\Rightarrow$ **cancel each other when N is large**

$\Rightarrow$ **all outputs identical regardless of routing!**

Product-space synthesis (v2.0 — CORRECT)

Compute the **full weight matrix** first:

$$\bar{W}_t = \sum_{i=1}^N A_{i,t} \cdot W_{i,t}^{\text{up}} W_{i,t}^{\text{down}}$$

No cross-terms. Then recover LoRA factors via truncated SVD (rank $r = 64$):

$$\bar{W}_t \xrightarrow{\text{tSVD}} U_t, S_t, V_t^T \Rightarrow \begin{cases} W_{\text{synth}}^{\text{up}} = U_t \sqrt{S_t} \\ W_{\text{synth}}^{\text{down}} = \sqrt{S_t} V_t^T \end{cases}$$

Why SVD? $\bar{W}_t$ is (1280×1280) , full-rank. SDXL only accepts LoRA format $(W^{\text{up}}, W^{\text{down}})$ at rank 64. tSVD finds the **best rank-64 approximation**.

Verified

Oracle test (GT=1.0): $\cos(W_{\text{synth}}, W_{\text{GT}}) = 0.9998$

SVD stability challenges (fp16 training)

- 1 **fp16 accumulation overflow:** Summing $N = 109$ fp16 matrices $\Rightarrow$ catastrophic cancellation
- 2 **cuSOLVER silent hang:** GPU SVD hangs on ill-conditioned matrices (no exception raised)
- 3 **NaN propagation:** fp16 softmax can produce NaN on extreme logits; NaN + noise = NaN

Final fix (all three)

- Accumulate $\tilde{W}_t$ in float32 regardless of autocast
- `nan_to_num(W, A)` before SVD
- SVD on **CPU (LAPACK)** — always terminates
- Cast U, S, V back to dtype for injection

Runtime cost

SVD call	5 ms (CPU)
80 SVDs / step	400 ms
Full forward step	5–8 s
Overhead	~8%

Matrices: $(d_{\text{out}} \times d_{\text{in}})$, e.g. (1280×1280)
Rank kept: $r = 64$ (out of min-dim)

Injection into SDXL

At inference: for each style block layer t :

$$W'_t = W_t + \alpha \cdot W_{\text{synth},t}^{\text{up}} W_{\text{synth},t}^{\text{down}}$$

α controls "how much style". See §Alpha calibration.

Why SVD? From Full Weight Matrix to LoRA Format

$$\bar{W}_t = \sum_i A_i W_i^{\text{up}} W_i^{\text{down}}$$

$(d_{\text{out}} \times d_{\text{in}})$, e.g. 1280×1280 , rank $\approx d_{\text{in}}$

full-rank — cannot inject as LoRA

LAPACK SVD (CPU)

U_t

$d_{\text{out}} \times d_{\text{in}}$

S_t (sorted)

d_{in} values, ↓

V_t^T

$d_{\text{in}} \times d_{\text{in}}$

keep top $r=64$

U_r

$d_{\text{out}} \times 64$

S_r

top-64

V_r^T

$64 \times d_{\text{in}}$

The problem with $\bar{W}_t$

- After synthesis, $\bar{W}_t$ is a full matrix: $(d_{\text{out}} \times d_{\text{in}})$
- SDXL's LoRA injection API requires **two small factors**: rank- r LoRA pair $(W^{\text{up}}, W^{\text{down}})$
- We can't inject $\bar{W}_t$ directly without replacing the weight — this would break checkpointing, memory, and the frozen-backbone property

What truncated SVD does

- Decomposes $\bar{W}_t = USV^T$ (full SVD)
- Singular values S : sorted by energy — most style information is in the top-64
- Keep only top $r = 64 \rightarrow$ lossy compression that matches B-LoRA's own rank
- Splitting $\sqrt{\epsilon}$ symmetrically gives

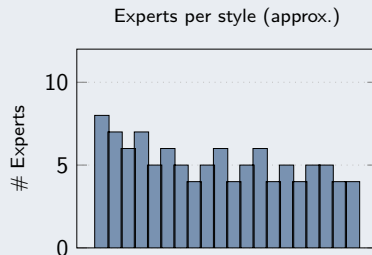
WikiArt (training images)

- **31,949 images** across 20 artistic styles
- Styles: Baroque, Realism, Impressionism, Cubism, Expressionism, Romanticism, ...
- Labels from `wikiart_label_map.json`
- Capped at 500 images/style for balance

B-LoRA Zoo (expert pool)

- **109 B-LoRA checkpoints** from `blora_zoo/bloras/`
- 20 WikiArt style categories
- Each: single-style SDXL LoRA, rank 64
- Features cached in `lora_features_cache.pt` (5.8 GB)

Style distribution



Approximately 5–8 experts per style

Stage 2 pool sampling

At each step: sample $N \in [5, 20]$ experts **excluding GT** from pool.
Encoder must find the GT-compatible style from negatives.

Two-Stage Training Pipeline

CLIP frozen Pool frozen

Stage 1 — Routing

Loss: **Cross-Entropy** (CE)

$$\mathcal{L} = - \sum_t \log A_{gt,t}$$

Train encoder to route WikiArt image to correct B-LoRA expert

15,000 steps LR=3e-4
Batch = 8 Pool size 5-20

checkpoint

CLIP, UNet, Pool frozen

Stage 2 — Denoising

Loss: **LDM + Entropy reg.**

$$\mathcal{L} = \mathcal{L}_{\text{LDM}} - \lambda \mathcal{H}(A)$$

Fine-tune encoder so that the *synthesised LoRA* reduces SDXL denoising error

8,000 steps LR=5e-5
fp16 mixed precision

Trains fast (~2h)
No SDXL needed

Full SDXL forward
Slow (~24h)

Stage 1: Cross-Entropy Routing Loss

Loss formulation

Attention $A \in \mathbb{R}^{N \times T \times r}$, GT index g :

$$\bar{A}_{i,t} = \frac{1}{r} \sum_j A_{i,t,j} \in \mathbb{R}^{N \times T}$$

For each of $T = 80$ adapter pairs, treat it as a classification problem: which of N experts should be activated?

$$\mathcal{L}_{\text{CE}} = -\frac{1}{T} \sum_{t=1}^T \log \bar{A}_{g,t}$$

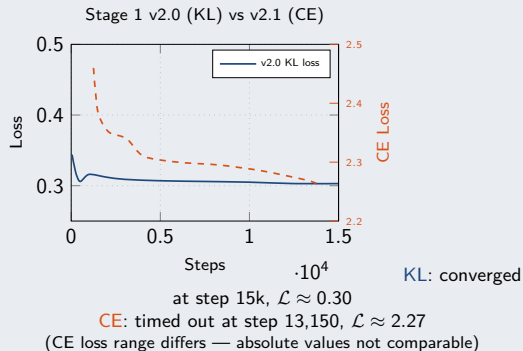
Gradient flows back through softmax into encoder weights.

Why CE over KL (v2.1 improvement)

KL mode (v2.0): targets from CLIP similarity between images and LoRA-generated images — CLIP clusters by *subject*, not style $\Rightarrow$ noisy soft labels

CE mode (v2.1): hard one-hot GT label from WikiArt category annotation $\Rightarrow$ unambiguous style supervision

Training curve (Stage 1)



Stage 2: LDM + Entropy Regularisation

Loss formulation

For a training image x with prompt p and pool of N experts:

$$\mathcal{L}_{\text{LDM}} = \mathbb{E}_{t, \epsilon} [\|\epsilon - \epsilon_{\theta}(\tilde{x}_t, t, p; W_{\text{synth}})\|_2^2]$$

Entropy regulariser (maximises routing diversity):

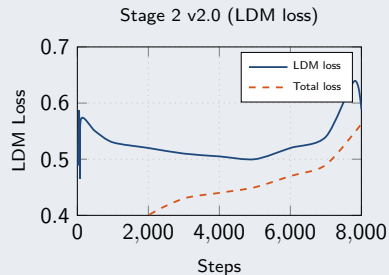
$$\mathcal{H}(A) = - \sum_{i, t, j} A_{i, t, j} \log A_{i, t, j}$$

Total loss:

$$\mathcal{L} = \mathcal{L}_{\text{LDM}} - \lambda(s) \mathcal{H}(A)$$

$\lambda(s)$: linearly decays $0.1 \rightarrow 0.01$ over training
(warm regularisation, then consolidate)

Training curve (Stage 2 v2.0)



completed 8,000 steps
LDM loss oscillates (stochastic batches)
Entropy term keeps routing diverse early

Stage 2

The Alpha Calibration Problem

Critical discovery

All early inference experiments produced **identical** images regardless of routing. Root cause:

$$\|\Delta W_{\text{synth}}\| \approx 16 \quad \text{vs} \quad \|W^{\text{up}} W^{\text{down}}\|_{\text{real B-LoRA}} \approx 50$$

At $\alpha = 1.0$: injected signal is **only 32% of a real B-LoRA** — below SDXL's visual threshold.

Alpha sweep findings

α	Observation
0.5	Identical to vanilla
1.0	Identical to vanilla
2.0	Visible style transfer
3.0	Style + minor distortion
5.0	Significant distortion

Sweet spot: $\alpha = 2.0$

Query: "A Baroque painting"

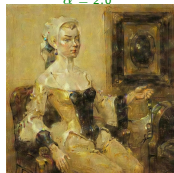
Vanilla



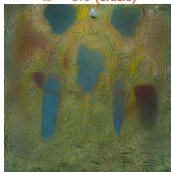
$\alpha = 1.0$



$\alpha = 2.0$

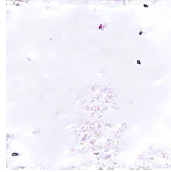


no injection
 $\alpha = 3.0$ (oracle)



content leakage

below threshold
 $\alpha = 5.0$



severe distortion

sweet spot
Ref B-LoRA (GT)



oracle ($\alpha = 1.0$)

Routing Analysis: Non-Uniform Attention Heatmap

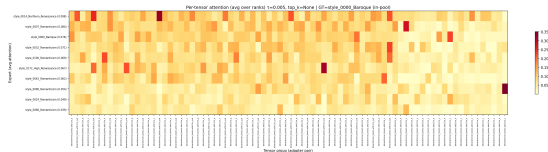
What the heatmap shows

Each row = one of $T = 80$ adapter pairs

Each column = one of N pooled experts

Colour = attention weight $A_{i,t}$ (avg over rank)

- Non-uniform: weights range 0.05–0.40+
- Structured per-layer pattern
- Confirms: **routing is working**

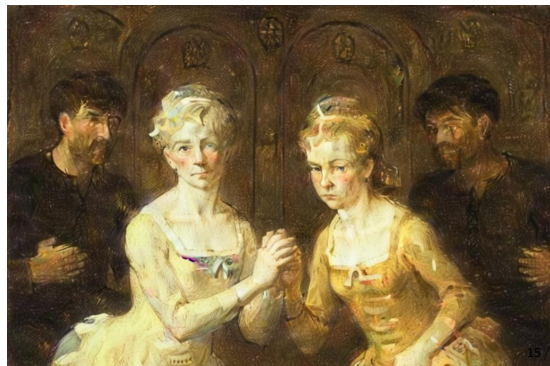


Heatmap: rows = 80 adapter pairs, cols = N experts.
Non-uniform weights confirm routing is active.

Top-5 experts (S1v2.1, Baroque query)

Rank	Style	ID
1	N. Renaissance	style_0014
2	Romanticism	style_0007
3	Baroque	style_0000
4	Romanticism	style_0032
5	Romanticism	style_0100

Baroque appears at rank 3. Top-2 are Renaissance/Romantic — all share dark palette & chiaroscuro with Baroque.
Routing is **semantically adjacent**, not exact label match.



Inference Sweep Setup

Experimental protocol

- **4 styles** tested: Baroque, Cubism, Expressionism, Impressionism
- **2 query modes**: pool (random WikiArt images) and wikiart (style-labeled images)
- **8 routing configurations** per style/query:
 - ps_t0.005_knone: sharp temp, all experts, product-space
 - ps_t0.005_k1: sharp temp, top-1 only (hard routing)
 - ps_t0.05_k*: moderate temp variants
 - ps_neutral: neutral temperature
 - ref_blora: reference ground-truth B-LoRA (oracle)
- **5 images per config = 80 total images per checkpoint**
- Inference: SDXL, 50 DDIM steps, $\alpha = 2.0$

Checkpoints evaluated

Checkpoint	Stage	Loss	Steps	Status
stage1_v2	S1	KL	15,000	✓
stage1_v21	S1	CE	13,150	✓
stage2_v2	S2	LDM+KL	8,000	✓
stage2_v21	S2	LDM+CE	pending	training...

Inference Results: Stage 1 v2.1 (CE) — Baroque

Pool query (random WikiArt) Prompt: "A Baroque painting" $\tau = 0.005$, $k = \text{None}$, $\alpha = 2.0$

Query image



WikiArt query (Baroque)

MoE synthesised



Synth LoRA output

Top-1 routed expert



N. Renaissance (rank 1)

Top-1 expert is Northern Renaissance (dark palette/chiaroscuro — stylistically adjacent to Baroque). Baroque appears at rank 3.

Inference Results: Stage 2 v2.0 (LDM+KL) — Baroque

Pool query Prompt: "A Baroque painting" $\tau = 0.005$, $k = \text{None}$, $\alpha = 2.0$

Query image



WikiArt query (Baroque)

MoE synthesised



Stage 2 output

Top-1 routed expert



Early Renaissance (rank 1)

Stage 2 routes to Early Renaissance as top-1 (stylistically adjacent). LDM fine-tuning shifts routing vs Stage 1 (Northern Renaissance top-1).

Inference Results: S1 v2.1 — Cubism

Pool query Prompt: "A Cubism painting" $\tau = 0.005$, $k = \text{None}$, $\alpha = 2.0$

Query image



WikiArt query (Cubism)

MoE synthesised



Synth LoRA output

Top-1 routed expert



Naïve Art / Primitivism (rank 1)

Cubism query routes to Naïve Art Primitivism (similar geometric flatness). Cubism itself appears at rank 5.

Inference Results: S1 v2.1 — Impressionism

Pool query Prompt: "An Impressionism painting" $\tau = 0.005$, $k = \text{None}$, $\alpha = 2.0$

Query image



WikiArt query (Impressionism)

MoE synthesised



Synth LoRA output

Top-1 routed expert



Impressionism (rank 1) ✓

Impressionism correctly retrieves Impressionism as top-1. Post-Impressionism at ranks 3–4 (adjacent style cluster).

Inference Results: S1 v2.1 — Expressionism

Pool query Prompt: "An Expressionism painting" $\tau = 0.005$, $k = \text{None}$, $\alpha = 2.0$

Query image



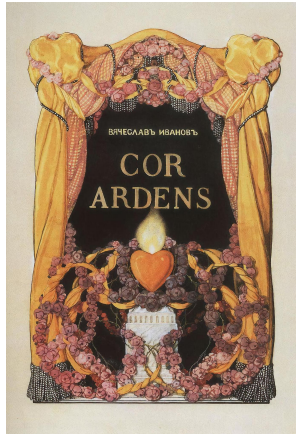
WikiArt query (Expressionism)

MoE synthesised



Synth LoRA output

Top-1 routed expert



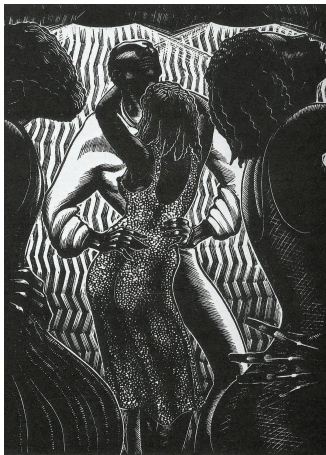
Symbolism (rank 1)

S1 routes to Symbolism (dark expressive brushwork, emotionally charged — adjacent to Expressionism). Expressionism itself at rank >5.

Inference Results: S2 v2.0 — Expressionism (S1 vs S2 Routing Shift)

Same query Prompt: "An Expressionism painting" $\tau = 0.005$, $k = \text{None}$, $\alpha = 2.0$

Query image



WikiArt query

MoE synthesised



Stage 2 output

Top-1 routed expert



Abstract Expressionism (rank 1) ↑

Routing Pattern vs Generated Image (4 Styles)

Baroque

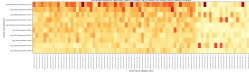


Routing (80 layers $\times$ N experts)



Generated: top-1=N.Renaissance

Cubism

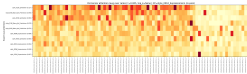


Routing (80 layers $\times$ N experts)



Generated: top-1=Naïve Art

Expressionism



Routing (80 layers $\times$ N experts)

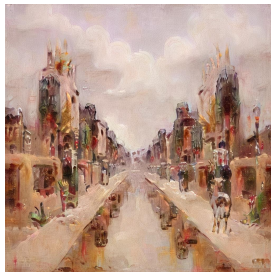


Generated: top-1=Symbolism

Impressionism



Routing (80 layers $\times$ N experts)



Generated: top-1=Impressionism ✓

Each style produces a distinct routing fingerprint

Top row: attention heatmap (rows=80 adapter pairs, cols=N experts). Bottom row: generated image at $\alpha = 2.0$. Impressionism shows a concentrated hot-spot (direct hit); others show dispersed patterns (adjacent-style routing).

Temperature Effect: Sharp vs Uniform Routing

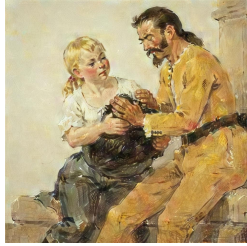
$\tau = 0.005, k = 1$
Hard routing



$\tau = 0.005, k = \text{None}$
Sharp+soft



$\tau = 0.5, k = \text{None}$
Uniform blend



Temperature τ

Low $\tau \Rightarrow$ winner-take-all

High $\tau \Rightarrow$ uniform over N experts

Default: $\tau = 0.005, k = \text{None}$
(sharp, all experts contribute)

Routing Heatmaps: All 4 Styles (S1 v2.1)

Baroque



Top-1: N. Renaissance

Cubism



Top-1: Naïve Art

Expressionism



Top-1: Symbolism

Impressionism



Top-1: Impressionism ✓

Distinct routing pattern per style

Rows = 80 adapter pairs; cols = N pooled experts. Impressionism: direct-hit top-1. Baroque/Cubism/Expressionism: semantically adjacent top-1.

Multi-Style Results Gallery (S1 v2.1)

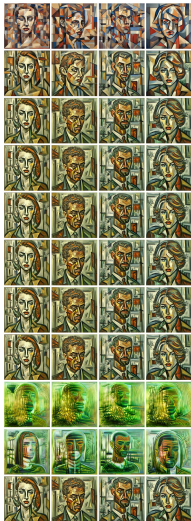
Baroque

High Resolution: 1024x1024, 16x16, 16x16, 16x16



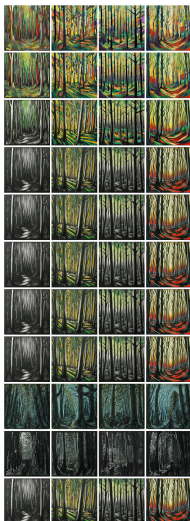
Cubism

High Resolution: 1024x1024, 16x16, 16x16, 16x16



Expressionism

High Resolution: 1024x1024, 16x16, 16x16, 16x16



Each grid: rows = routing configs ($\tau \in \{0.005, 0.05, 0.5\}$, $k \in \{1, \text{None}\}$) $\times$ cols = query images. Bottom row: reference B-LoRA (GT expert, oracle).

Model Variants Summary

Model	Stage	Loss	Steps	Synth.	α	Status
v1.0 baseline	S1	KL	5k	naïve avg	1.0	obsolete
S1 v2.0 (KL)	S1	KL	15,000	product-sp.	2.0	✓ done
S2 v2.0 (KL+LDM)	S2	KL+LDM	8,000	product-sp.	2.0	✓ done
S1 v2.1 (CE)	S1	CE	13,150	product-sp.	2.0	✓ done
S2 v2.1 (CE+LDM)	S2	CE+LDM	—	product-sp.	2.0	training...
<i>Ref B-LoRA</i>	—	—	—	GT LoRA	1.0	Oracle

Key design differences

- **v2.0 vs v2.1**: training loss (KL vs CE); v2.1 uses hard one-hot GT, avoids noisy CLIP-similarity targets
- **S1 vs S2**: Stage 1 trains purely on routing; Stage 2 additionally closes the loop via SDXL denoising loss

4-way comparison pending

Once S2 v2.1 training completes, we run the same 80-image sweep on all 4 checkpoints for a direct quality comparison.

Open Issue: Prompt Mismatch

Critical unresolved problem

B-LoRA experts were trained with trigger token "A [v]".

Our inference sweeps use *free-text* prompts like "A Baroque painting".

Context

B-LoRA training (original work):

- Each expert fine-tuned on 30 images with prompt "A [v]"
- [v] is a **special learned token** embedded into SDXL's textual cross-attention
- At inference, "A [v]" activates the specific style representation

Our setup

- We use prompts like "A Baroque painting", "A painting in Cubist style", etc.
- SDXL text embeddings for these are *completely different* from the token embedding [v]
- The LoRA weights may not activate optimally without

Impact

- Style fidelity may be degraded even with correct routing
- Proper controlled experiment should use "A [v]" as prompt for all generated images
- Current results are a **lower bound** on achievable quality

Proposed fix

- 1 Run a parallel sweep using "A [v]" for all generations
- 2 Compare: free-text prompt vs trigger token
- 3 This isolates the prompt effect from the routing quality

Challenge 1: $O(N^2)$ cross-term bug

Symptom: All inference outputs identical regardless of routing.

Root cause: Naïve LoRA averaging introduces $O(N^2)$ cross-terms that cancel all style signal when N is large.

Fix: Product-space synthesis — compute $\bar{W}_t = \sum A_i W_i^{\text{up}} W_i^{\text{down}}$, then tSVD.

Challenge 2: Alpha sub-threshold

Symptom: Correct routing, still identical images.

Root cause: $\|\Delta W_{\text{synth}}\| \approx 16$ vs real B-LoRA ≈ 50 ; at $\alpha = 1.0$ only 32% signal strength.

Fix: Use $\alpha = 2.0$; also set TARGET_NORM=32 for norm_match mode.

Challenge 3: SVD numerical instability

Three sub-issues:

- 1 fp16 accumulation of $\bar{W}_t$ causes underflow/overflow in $N=109$ sum $\Rightarrow$ Fix: accumulate in fp32
- 2 cuSOLVER silently hangs on ill-conditioned matrices (no exception) $\Rightarrow$ Fix: SVD on CPU (LAPACK)
- 3 fp16 softmax NaN poisons the accumulation; noise retry also NaN $\Rightarrow$ Fix: nan_to_num before SVD

Current status (job 765827 on ai12)

Step 875/8000 — running. LDM loss ≈ 0.55 (improving).

Note: ent=-0.000000 — entropy term not contributing. Attention distribution may be near-uniform (flat softmax). This is fix #5; all previous SVD/NaN crashes now resolved.

Next Steps

Immediate (days)

- 1 Confirm Stage 2 v2.1 (CE+LDM) training succeeds
- 2 Run inference sweep on S2 v2.1 ($\alpha = 2.0$)
- 3 **4-way comparison:** S1v2.0 / S2v2.0 / S1v2.1 / S2v2.1

Short-term (week)

- 1 Controlled experiment: "A [v]" prompt vs free-text
- 2 Qualitative evaluation using CLIP-style similarity metric
- 3 Ablation: temperature, top- k effect

Medium-term

- 1 Quantitative metrics: FID, LPIPS, style CLIP-similarity
- 2 Multi-style queries (interpolation between styles)
- 3 Out-of-distribution testing (styles not in zoo)
- 4 Potential: joint fine-tuning of encoder + minimal LoRA layers

Key open question:

Does Stage 2 LDM fine-tuning improve style quality over Stage 1 CE routing only?

What we built

- **MoELoRA v2**: per-tensor rank-level attention router over 109 B-LoRA style experts
- **LoRARankEncoder** (2.2M params): shared point-wise MLP encoding LoRA rows to key vectors
- **Product-space synthesis** + tSVD: correct, numerically stable multi-expert blending
- **Two-stage training**: CE routing (S1) + LDM denoising loop (S2)

Key findings

- Routing **is working** (non-uniform heatmaps, semantically relevant top- k)
- $\alpha = 2.0$ is the sweet spot for visible, non-distorted style injection
- CE loss (v2.1) gives cleaner routing targets than KL

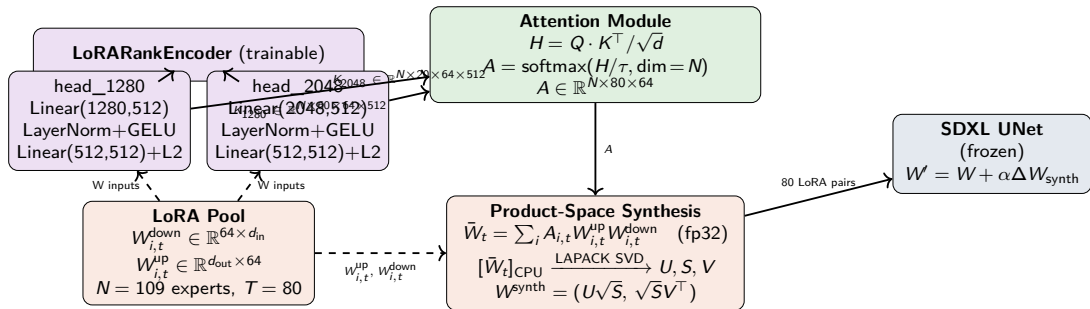
Open challenges

- Prompt mismatch: "A [v]" vs free-text
- Stage 2 SVD stability (fp16 training, 4 attempts)
- No quantitative evaluation yet

Completed experiments

- ✓ S1 v2.0 sweep: 80 images
- ✓ S1 v2.1 sweep: 80 images
- ✓ S2 v2.0 sweep: 80 images
- ✓ Alpha diagnostic: 12 configurations
- ... S2 v2.1: training in progress

Appendix: Complete Model Architecture



Appendix: Oracle Test — Synthesis Fidelity

Oracle routing experiment

Set attention weight 1.0 on GT expert, 0.0 on all others.
Product-space synthesis should recover the GT LoRA exactly.

$$A_{\text{oracle}} = e_g \Rightarrow \bar{W}_t = W_{g,t}^{\text{up}} W_{g,t}^{\text{down}}$$

SVD of a rank- r matrix:

$$W_{g,t}^{\text{up}} W_{g,t}^{\text{down}} \xrightarrow{\text{tSVD}} W_{\text{synth}}^{\text{up}}, W_{\text{synth}}^{\text{down}}$$

Measured result

$$\cos\left(W_{\text{synth}}^{\text{up}} W_{\text{synth}}^{\text{down}}, W_g^{\text{up}} W_g^{\text{down}}\right) = \mathbf{0.9998}$$

$\Rightarrow$ synthesis is numerically lossless at oracle.

(Small error from float32 SVD truncation at rank 64)



Oracle output ($\alpha = 3.0$): clear content leakage from top-1 expert confirms injection is working.

Appendix: Training Configuration

Stage 1 hyperparameters

	v2.0 (KL)	v2.1 (CE)
Loss	KL div	Cross-Entropy
Max steps	15,000	15,000
LR	3e-4	3e-4
Warmup	500	500
LR sched.	cosine	cosine
Batch size	8	8
Pool size	5–20	5–20
τ label	0.3	—
Grad clip	1.0	1.0

Stage 2 hyperparameters

Param	Value
Max steps	8,000
LR	5e-5
Warmup	200
Precision	fp16 mixed
λ_{ent}	0.1 $\rightarrow$ 0.01
Batch size	1
Pool size	5–20 (no GT)
LoRA α	1.0 (training)

Hardware

VALAR HPC · NVIDIA A100 GPU
SLURM scheduling · 24h time limit

Thank You

Questions?

Code: `MoLoRAs/lora_attention/`

Results: `/scratch/eyavuz21/lora_attention/`

Roadmap: `lora_attention/roadmap.md`